

Datathon Stream 2

Team: Tavish and William

6 September 2026

Approach

We used LightGBM on a large set of time-series features specifically extracted from the data using PySpark.

Data preparation

We combined the train and test tables before building features, because lag features need the hours before a row to exist in the same table. Beyond that, preprocessing was minimal: we cast the numeric columns consistently, treated station as a categorical variable, and left missing values as they were for LightGBM to handle.

Features

Roughly 300 features in a few groups:

- Calendar: hour, day, month, plus sine/cosine versions so the model treats midnight and 1 a.m. as neighbours.
- Weather: wind direction turned into vectors, and relative humidity calculated from temperature and dew point.
- History: lags of PM10 going back 48 hours, shorter lags for the other pollutants, and rolling means, minimums, maximums and standard deviations.
- Same time readings: the pollutant and weather values at the target time.
- Trends: how much each measurement has changed over the last few hours.
- City-wide: the average across all stations each hour, so the model can tell "the whole city is polluted" apart from "this station is spiking".
- Interactions: mainly PM10 combined with humidity, since fine particles swell as the air gets more humid.

Validation

We validated by season rather than randomly, since random splits would cause data leakage. For each year we trained on everything up to 31 August and validated on September through February. We also down-weighted older training rows, as newer data is expected to be more accurate.

Model

Three LightGBM models, blended:

- a smaller, heavily regularised model (63 leaves);
- a larger one that picks up sharper detail (160 leaves);
- a copy of the first one, but trained on the log of the target, which works better on very high pollution hours.

They make different mistakes, which is why blending them helps. Blend weights were found by a simple grid search on out-of-fold predictions.

Post-processing

We checked whether a linear recalibration of the predictions helped, it did not. Each model was retrained under two random seeds and averaged. Predictions were clipped to a plausible range.

Results

PM10 was by far the strongest predictor, which makes sense since PM2.5 is part of PM10. Humidity interactions and the city-wide averages were also strong predictors.

Reproducing it

One script runs the entire model, from feature engineering through to writing `submissionFinal.csv`. It needs Python 3.12, PySpark, LightGBM, pandas and numpy (library versions in `requirements.txt`). Leave `FAST = False` as the fast mode runs a single fold for debugging and will not reproduce the submission. Seeds are 42 and 202. No intermediate files need saving; everything is rebuilt during the run.

Disclosure

Claude AI was used for implementation and debugging of code.